MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE -50° TO +50° C

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for a single-pointer, white-lighted, electrical resistance temperature indicator.

* 1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

* ABU-10/A Anti-reflective coating on coverglass per MIL-C-675.
* ABU-10A/A High efficiency anti-reflective coating on coverglass.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

	MIL-P-116	Preservation Method of.
*	MIL-C-675	Coating of Glass Optical Elements (Anti-reflection).
	MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
	MIL-I-7070	Indicator, Temperature, Electrical-Resistance, Hermetically Sealed, General Specification for.
	MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for.

Beneficial comments (recommendations, additions, deletions) and and pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packaging of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-831	Test Reports, Preparation of.
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft.
MS33678	Connector, Receptacle, Electric, Integral Mounting.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

FEDERAL AVIATION AGENCY

Technical Standard Order

TSO-C43	Temperature Indicators

(Copies of FAA documents may be obtained from the Federal Aviation Agency, Washington, D.C. 20533.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article - When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.4).

3.2 General specification. The requirements of MIL-I-7070 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

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3.3 Construction.

3.3.1 Case. The case shall conform to MS33639 for the 2 inch diameter size. The case length shall be 3.50 ±0.25 inch.

* 3.3.2 Dial. The dial shall conform to figure 1 and as specified herein. The indicator scale shall be as long as practicable and shall be not less than 1 3/4 inches as measured on the outer arc of the shortest graduation. The 0°C graduation shall be located on the horizontal centerline of the dial. The dial blank shall consist of a flat disc of plastic, metal, or other material that will permit compliance with the lighting and other requirements specified herein.

* 3.3.2.1 Coverglass coating. The reflecting glass surface of the ABU-10/A indicator shall be finished with a reflection reducing coating conforming to MIL-C-675. The ABU-10A/A indicator shall be finished with a high efficiency coverglass coating conforming to the following. The coatings shall withstand the environmental conditions specified herein.

	Reflectance tolerances	
Angle of incidence	Wavelength millimicrons	Percent reflectance
0°	450 - 675	0.6 Maximum
0°	425 - 700	0.5 Average
30°	450 - 625	1.0 Maximum
30°	425 - 700	0.5 Average

* 3.3.3 Pointer. Within a distance of 0.50 ±0.063 inch from the tip, the pointer width shall taper from 0.078 ±0.016 inch to 0.020 ±0.005 inch at the tip. The length of the pointer shall be such that the tip of the pointer extends into the shortest scale graduation by 0.019 ±0.016 inch.

3.3.3.1 Pointer overswing. The pointer overswing shall be not less than 3 percent of full scale angular deflection beyond each end of the scale.

* 3.3.4 Lighting. The dial and pointer shall be lighted in accordance with MIL-L-27160.

3.4 Dial to cover glass location. The maximum distance between the inner surface of the cover glass and the dial surface shall not exceed 0.188 inch.

3.5 Performance.

* 3.5.1 Vibration shall be in accordance with MIL-STD-810. Test Procedure I-3.2.4 and/or I-3.2.5.

3.6 Electrical receptacle. The MS33678-14S-5P hermetically sealed, electrical receptacle shall be provided and shall be wired as follows:

Pin	Connection
A	+28V dc
B	Resistance bulb connection
C	Ground and remaining bulb connection
D	5V lighting input
E	5V lighting ground

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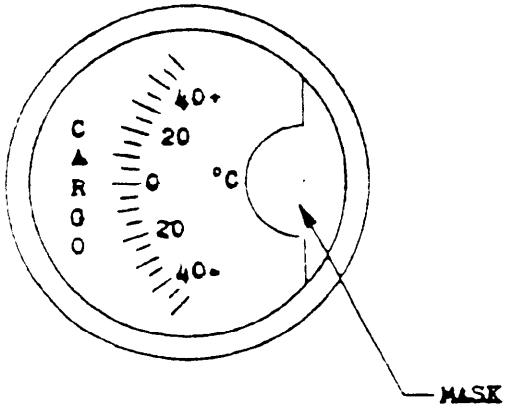


FIGURE 1. dial.

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3.7 Electronic components. Electronic components and application thereof shall conform to MIL-E-5400.

3.8 Power consumption. The power consumption of the indicator, exclusive of the lighting circuit, shall not exceed 2 watts.

3.9 Markings.

3.9.1 Dimensions and colors. The dimensions and colors of the markings shall be as follows. The colors shall be in accordance with FED-STD-595.

Marking	Height or length ±0.005	Width of line or graduation ±0.005	Color No.
All numerals	0.125	0.020	37875
Major graduations	0.156	0.031	37875
Minor graduations	0.093	0.020	37875
Lettering °C	0.125	0.020	37875
Cargo	0.125	0.020	37875
Background of dial and mask	-	-	37038

3.10 Weight. The weight of the indicator shall not exceed 1.0 pound.

3.11 Nameplate. A nameplate shall be permanently attached to the rear of the indicator case. The nameplate shall be made of metal. Metal cals shall not be used.

3.12 Reliability. The indicator shall have a mean-time-between-failure (MTBF) of 280 hours demonstrated at a 90 percent confidence level based on the chi-squared distribution.

3.13 TSO conformance. The indicator shall conform to the requirements of Federal Aviation Agency TSO-C43 for type I indicators.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are as follows:

a. First Article Inspection (see 4.2.1).

b. Quality Conformance Inspection (see 4.2.4).

* 4.2.1 First article tests. The first article tests shall consist of the first article tests of MIL-I-7070.

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* 4.2.2 Sampling for first article tests. Unless otherwise specified as soon as practicable after the award of the contract or order, the contractor shall submit three indicators to the first article tests as specified herein.

* 4.2.3 First article test report. The contractor shall prepare a report of the first article tests in accordance with MIL-STD-831. Two copies shall be furnished to the procuring activity.

* 4.2.4 Quality Conformance Inspection. The inspection and testing of the indicator shall be in accordance with MIL-I-7070 except as specified herein.

4.3 Test conditions.

4.3.1 Test voltage. Unless otherwise specified, the test voltage shall be 28.5 ±0.5V dc and the lighting circuit shall be energized with 4.5 ±0.5V during all tests.

4.3.2 Test tolerances. The tolerances specified in table 1, not enclosed in parentheses, are applicable only when the indicators are subjected to all tests in the sequence specified herein. The tolerances specified in table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4 Test methods.

4.4.1 Individual tests.

4.4.1.1 Scale error at room temperature. The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in table I shall not exceed the tolerances specified in table I.

4.4.1.2 Position error. The maximum change in indication shall not exceed 2° C.

4.4.1.3 Friction error. The maximum difference in reading before and after tapping shall not exceed 2° C.

4.4.1.4 Insulation leakage. The insulation resistance measured between any pin and the shell, shielding, or any metal part of the indicator shall be a minimum of 20 megs.

4.4.1.5 The indicator shall be subjected to the individual tests of MIL-L-27160.

4.4.1.6 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. Where a nitrogen-helium mixture is utilized as a filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of one atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

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TABLE I. Test tolerances.

Test point	Test resistance	Scale error at room temperature	Reduced voltage	Scale error at low temperature	Scale error at high temperature	Scale error at room temperature after open and short circuit	Scale error at room temperature after vibration failure
oC	Ohms	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC
-50	74.24	1.5(1.5)	1.0(1.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
-30	80.56	1.5(1.5)	1.0(1.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
-10	87.04	1.5(1.5)	1.0(1.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
0	90.38	1.0(1.0)	1.0(1.0)	2.0(2.0)	2.0(2.0)	1.5(1.5)	3.5(3.0)
+10	93.80	1.5(1.5)	1.0(1.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
+30	100.91	1.5(1.5)	2.0(2.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)
+50	108.39	1.5(1.5)	2.0(2.0)	3.0(3.0)	3.0(3.0)	2.0(2.0)	4.0(3.5)

Test point	Test resistance	Extreme low temperature exposure		Scale error at room temperature after high temperature exposure	Scale error with bulb lead resistance
		Scale error at low temperature	Scale error at room temperature after low temperature exposure		
oC	Ohms	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC	Tolerance ±oC
-50	74.24	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
-30	80.56	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
-10	87.04	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
0	90.38	4.5(2.0)	3.5(1.0)	3.5(1.0)	1.5(1.5)
+10	93.80	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
+30	100.91	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)
+50	108.39	5.5(3.0)	4.0(1.5)	4.0(1.5)	1.5(1.5)

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4.4.2 Sampling plan A.

4.4.2.1 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5V at the test points specified in table I. The indications at the reduced voltage shall not differ from the indications with 28.5V applied by more than the tolerance specified in table I.

4.4.2.2 Scale error at low temperature. The scale errors at low temperature shall not exceed the tolerances specified in table I.

4.4.2.3 Scale error at high temperature. The scale errors at high temperature shall not exceed the tolerances specified in table I.

4.4.2.4 Magnetic effect. The deflection of the compass shall not exceed 5°.

4.4.2.5 Resistance bulb current. A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 ma.

4.4.2.6 Open circuit and short circuit. After open and short circuit, the scale errors at room temperature shall not exceed the tolerances specified in table I.

4.4.2.7 Vibration. The indicator shall be subjected to vibration in accordance with MIL-STD-810. Pointer oscillation shall not exceed 2° C double amplitude. Pointer variation shall not exceed 2° C. The test shall be conducted with power applied and with input resistance to provide an indication of 0° C. The resonance and cycling portions of this test shall be conducted through the specified range at vibratory accelerations not exceeding those shown for curve B. When pointer oscillation and pointer variation are noted, the vibration frequency shall be 5 to 500 cps at double amplitudes that will produce a vibration level of 1.5g maximum.

4.4.2.8 Scale error at room temperature after vibration. After vibration, the scale errors at room temperature shall not exceed the tolerances specified in table I.

4.4.2.9 The sampling plan A tests specified in MIL-L-27160 shall be conducted.

4.4.3 Sampling plan B.

4.4.3.1 Response time. The response time shall not exceed 3 seconds.

4.4.3.2 Damping factor. The damping factor shall be not less than 1.6.

4.4.3.3 Power consumption. The power consumption shall not exceed 2 watts.

4.4.3.4 Bulb lead resistance effect. The indicator shall be connected as shown on figure 2 and shall be subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in table I.

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4.4.3.5 Extreme low temperature exposure. After low temperature exposure, the scale errors at low temperature and the scale errors at room temperature shall not exceed the tolerances specified in table I.

4.4.3.6 High temperature exposure. After high temperature exposure, the scale errors at high temperature and the scale errors at room temperature shall not exceed the tolerances specified in table I.

4.4.3.7 Salt spray. The indicator shall be subjected to and shall meet the requirements of the sealing test following salt spray exposure. The insulation resistance shall be a minimum of 20 megs.

4.4.3.8 The sampling plan B tests specified in MIL-L-27160 shall be conducted.

4.4.3.9 Thermal shock. No evidence of moisture penetration or damage to the inclosure shall result from this test. Following this test, the indicator shall be resubjected to and shall meet the requirements of the sealing test specified in 4.4.1.6. Fogging of the coverglass during any part of this test shall be cause for rejection.

4.4.3.10 Reliability. The first article test samples shall first be subjected to the scale error test and then to cycles consisting of 4 hours of operation with power and resistance applied and 1 hour with power removed. The cycles shall be conducted at the following ambient temperatures, $\pm 2^{\circ}C$, in the sequence specified: 4 hours at $0^{\circ}C$, 1 hour off; 4 hours at room temperature, 1 hour off; 4 hours at $50^{\circ}C$, 1 hour off. During the operational portion of the cycles, the resistance shall be so varied as to produce an indication cycle corresponding to a change from -40° to $+40^{\circ}$ and return to $-40^{\circ}C$. This shall constitute 1 indication cycle which shall be repeated at a rate of 6 ± 1 cpm until 75,000 indication cycles have been completed. The continuity of this test may be interrupted for weekend shutdown. At the conclusion of this test, the indicators shall be subjected to the scale error at room temperature and the friction error tests. The scale errors at the respective test points shall not exceed 150 percent of the scale errors prior to this test, and the friction error shall not exceed $3^{\circ}C$. Reliability of the equipment will be verified when the indicators have passed all tests specified herein.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation applications. Preservatives shall not be used.

5.1.1.4 Unless otherwise specified (see 6.2) each indicator shall be individually unit packed in accordance with method 1A of MIL-P-116.

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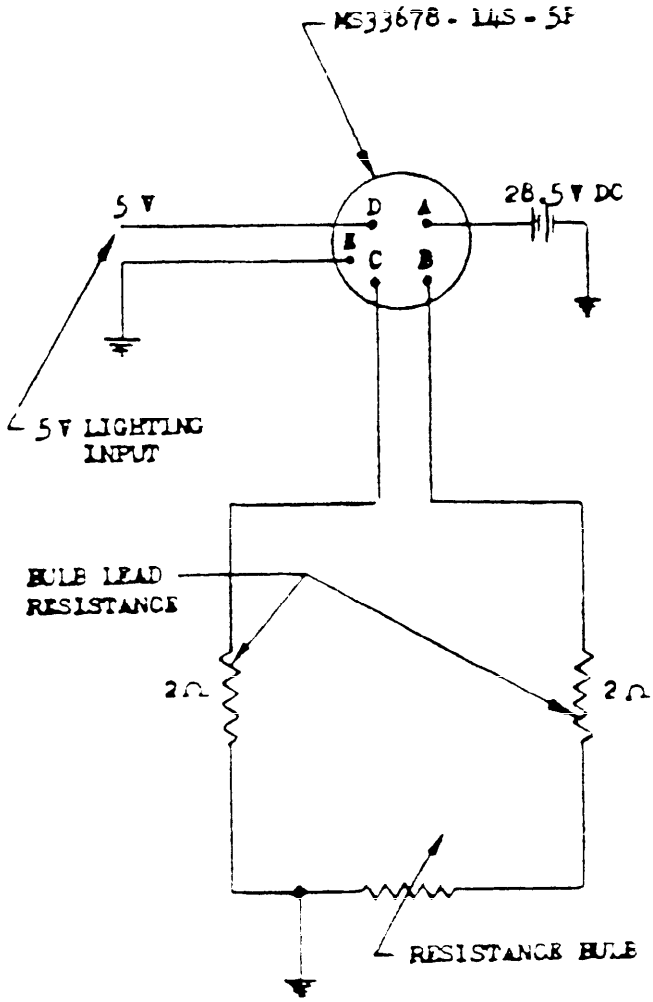


FIGURE 2. wiring diagram.

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5.1.2 Level C. The level C preservation for indicators shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial/industrial preservation of indicators shall be in accordance with the requirements of ASTM D 3951-82.

5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2).

5.2.1 Level A. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level A protection.

5.2.2 Level B. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level B protection.

5.2.3 Level C. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level C protection.

5.2.4 Commercial. The packaged indicators shall be packed in accordance with the requirements of ASTM D 3951-82.

5.3 Marking.

5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951-82.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating temperature, such as cargo compartment temperature, within a range of -50° to $+50^{\circ}$C. The indicators may be used on instrument panels requiring case mounting by application of an MS28042-1 instrument mounting clamp.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of indicator required (see 1.2).

c. Level of packaging and packing desired.

d. When sampling plan B tests will not be conducted.

6.3 Definitions. Definitions will be as specified in MIL-I-7070.

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6.4 First Article - When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-27665	Indicator, Temperature, Electrical Resistance -50° to +50°C

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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A leak is defined as the detection of more than 1 Vppm at the effluent side of

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